Roadmap: measured so far, planned next

Goal: about 1B parameters, text not worse than a matched autoregressive baseline.

COMPLETED

TinyWorld

Synthetic grammar
Codec, generator, AR
Latency and quality

Sudoku

E6 to E20, two seeds
Easy 98.1%, medium 83.4%
Hard boards still behind

Checkpoint port

Runs Kumbong et al.
TinyStories and Sudoku
Agrees with PyTorch, tested

TinyStories

E21 to E24, two seeds
Lower loss, worse text
Best NCM perplexity 3x AR

PLANNED, IN ORDER

01 Close the gap at small scale

Best NCM perplexity is about 3x AR; corruption and loss-weighting ablations

02 Prompt-conditioned training

Official XSum and WMT14 codecs
Red-fuchs/ncm-modular, issue #3

03 Mixed-precision throughput

Measure speed before scaling up

04 Natural-text corpus

Several billion tokens

05 100M model, about 2B tokens

The go / no-go against AR

06 300M, then about 1B

Only if NCM is close to AR at matched compute

Planned steps are plans, not results.

No delivery dates.